

Cloud-Native Telecommunications Infrastructure

Chapter 1. Containerized Network Functions (CNFs)

Modern telecommunications operators are transitioning from Virtual Machines (VNFs) to Containerized Network Functions (CNFs). CNFs leverage microservices architectures orchestrated by Kubernetes, enabling automated lifecycle management, zero-downtime rolling deployments, and optimized compute footprint.

Unlike traditional enterprise workloads, telecom user plane workloads demand deterministic low latency and extreme packet throughput.

Chapter 2. High-Performance Networking Acceleration

To achieve wire-speed processing in containerized environments, CNFs utilize specialized kernel-bypass acceleration technologies:

1. Single Root I/O Virtualization (SR-IOV): Allows a single physical PCIe network interface card (NIC) to appear as multiple virtual devices (VFs), which can be passed directly into container pods.
2. Data Plane Development Kit (DPDK): Provides user-space poll-mode drivers that bypass the Linux kernel network stack, eliminating interrupt processing and context-switching overhead.
3. Express Data Path (XDP) and eBPF: Enables programmable in-kernel packet filtering and steering at the device driver layer before socket buffers are allocated.

Chapter 3. Kubernetes CNI Plugins in Telco Deployments

Standard Kubernetes clusters provide a single network interface per pod using a Container Network Interface (CNI) plugin like Flannel or Calico. However, telecom CNFs require multiple network interfaces: one for cluster management and control plane traffic, and one or more high-performance interfaces for user plane packet forwarding.

Multus CNI acts as a meta-plugin that attaches multiple network interfaces to a single Kubernetes pod. Secondary interfaces can be backed by Macvlan, IPvlan, or SR-IOV device plugins.

Chapter 4. Zero-Trust Security and Observability

Telecom cloud infrastructure operates under a Zero-Trust Architecture (ZTA). Inter-service communication is secured via mutual Transport Layer Security (mTLS) with cryptographic identity certificates issued by SPIFFE/SPIRE agents.

Observability is provided through OpenTelemetry standards:

- Metrics collection via Prometheus pull endpoints.
- Distributed tracing using W3C TraceContext propagation across asynchronous message queues.
- Real-time packet telemetry collected via eBPF probes for anomaly detection.